Carla Nathaly Villacís Núñez

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Education

University of Michigan, Ann Arbor, MI

Ph.D. in Mechanical Engineering and Scientific Computing 2025

- Dissertation: *Elucidating rotator cuff tendon tear growth mechanisms with full volume strains and data-driven modeling*
- Advisor: Professor Ellen Arruda

M.S.E. in Mechanical Engineering 2020

- Independent research project: *Asymmetric multi-bevel needle tip geometry for accurate needle insertion*
- Advisors: Professor Albert Shih and Professor Dian-Ru Li

Certificate in Innovation & Entrepreneurship 2020

Universidad de las Fuerzas Armadas - ESPE, Quito, Ecuador

Mechanical Engineer (Valedictorian) 2014

- Thesis: *Design and construction of an experimental prototype of a PET bottle perforating machine for the company NEUMAC S.A.*
- Advisors: Professor José Pérez and Professor Milton Acosta

Research Experience

Schmidt AI in Science Postdoctoral Research Fellow 2025–Present

Michigan Institute for Data & AI in Society, Ann Arbor, MI

Mentors: Professor Vikram Gavini and Professor Xun Huan

- Developing data-driven digital twin algorithms for rotator cuff tendons integrating full volume experimental data with computational modeling to predict injuries and optimize repair strategies
- Collaborating with Professor Talia Moore on the development of a diffusion algorithm to discover hourglass geometry from stress fields coupled with force-angle curves
- Collaborating with Professor Jacinto Ulloa on the generation of a phase field inference algorithm for soft materials with anisotropy
- Wrote a Community Initiatives Funds proposal to host a workshop on translational research approaches beyond the clinical field, in collaboration with Dr. Daniela Andrade Damasceno and Dr. Claire Baker
- Created a predictive model for alcohol use disorder that incorporated uncertainty quantification, in collaboration with Professor Dimah Dera and Professor Chenlan Wang
- Volunteered to judge posters, assist in note taking, and moderate discussions during AI Summit

Doctoral Researcher 2020–2025

University of Michigan, Ann Arbor, MI

- Established an MRI-based experimental-computational framework to study rotator cuff tear progression combining imaging, solid mechanics, and data-driven system identification
- Created scalable computational pipelines to infer parsimonious constitutive behavior in the rotator cuff, applicable to broader biological systems
- Pioneered 3D strain measurement and 3D inverse modeling algorithms in the rotator cuff, producing 10 awards, 2 peer-reviewed publications, 10 conference talks and 5+ posters
- Designed an adjustable experimental setup to fit a variety of bone shapes and sizes, enabling strength measurement of repaired tendon-bone complexes
- Initiated a collaboration with Dr. Carlos Urrego to study diabetic mouse bones through finite element analysis, which resulted in a multidisciplinary manuscript
- Wrote two pilot grant proposals to analyze torn and repaired tendons with full volume strains

Independent Researcher 2019–2020
University of Michigan, Ann Arbor, MI

- Conducted finite element simulations of biological structures to study evolutionary biomechanics
- Prototyped and built a vision-guided system for real-time control of needle tip micro-grinding, achieving 50-micron accuracy and reducing deflection during biopsy procedures

Rheology Laboratory Analyst and Researcher 2017–2018
Universidad de las Fuerzas Armadas - ESPE, Quito, Ecuador

- Refined a proposal on how to leverage Ecuadorian ancestral knowledge to develop membranes of silk fibroin and chitosan from an electrospinning technique

Peer-Reviewed Publications

J1. Villacís Núñez, C. N., Srivastava, S., Scheven, U., Bedi, A., Garikipati, K., & Arruda, E. M. (2026). Constitutive parameter inference using physics-informed full volume inverse modeling of intact and torn rotator cuff tendons. *Journal of the Mechanics and Physics of Solids*. 106668 (214). https://doi.org/10.1016/j.jmps.2026.106668

J2. Hernandez Lamberty, M. A., **Villacís Núñez, C. N.**, Scheven, U., Grant, J. A., Arruda, E. M., & Coleman, R. M. (2025). Preliminary evaluation of full volume strain measurement in patellar cartilage following osteochondral allograft transplantation using magnetic resonance imaging. *Frontiers in Bioengineering and Biotechnology*. 13, 1701592. https://doi.org/10.3389/fbioe.2025.1701592

J3. Villacís Núñez, C. N., Scheven, U., Bedi, A., & Arruda, E. M. (2025). Tear growth mechanisms in high-grade bursal-sided partial thickness tears in the rotator cuff measured with full volume magnetic resonance imaging methods. *Acta Biomaterialia*. 203 (2025), 438-450. https://doi.org/10.1016/j.actbio.2025.07.038

J4. Villacís Núñez, C. N., Poli, A., Stanifer, E., Mao, X., & Arruda, E. M. (2023). Fractional topological solitons in nonlinear viscoelastic ribbons with tunable speed. *Extreme Mechanics Letters*. 61 (2023), 102027. https://doi.org/10.1016/j.eml.2023.102027 (**Journal cover page**)

J5. Kennedy, C. S., **Villacís Núñez, C. N.**, Poli, A., Vega-Soto, E., Arruda, E. M., & Larkin, L. M. (2023). Engineered tissue graft for repair of injured infraspinatus rotator cuff tendon. *Tissue Engineering Part A*. 29(17-18), 471-480. https://doi.org/10.1089/ten.TEA.2022.0196

J6. Villacís Núñez, C. N., Ray, A. P., Cooper, K. L., & Moore, T. Y. (2022). Metatarsal fusion resisted bending as jerboas (Dipodidae) transitioned from quadrupedal to bipedal. *Proceedings of the Royal Society B*. 289 (1984), 20221322. https://doi.org/10.1098/rspb.2022.1322

J7. Villacís, N., Gualavisí, M., Narváez-Muñoz, C., Carrión, L., Loza-Matovelle, D. & Naranjo, F. (2017). Additive manufacturing of a rheological characterized cement-based composite material. *Proceedings of the 2017 European Conference on Electrical Engineering and Computer Science (EECS)*. 326-331, https://doi.org/10.1109/EECS.2017.67

Manuscripts Under Review

R1. Urrego, C., Raigosa, J. P., Jomaa, S. W., Quint, A. G., **Villacís Núñez, C. N.**, Sexton, B. S., Kohn, D. H. (2026). Compromised femoral architecture associated with spatially disrupted osteocyte lacuno-canalicular network leads to stress redistribution in type 2 diabetic female mice. [Submitted to JBMR Plus]

Manuscripts In Preparation

P1. Villacís Núñez, C. N., Scheven, U., Bedi, A., Arruda, E. M., Gavini, V., & Huan, X. (2026). Digital twin of a rotator cuff tendon surrogate: A proof of concept validating physical-digital communication and model refinement. [In preparation]

Invited Talks

IT1. Villacís Núñez, C. N. Toward Digital Twins for Soft Tissue Injury and Repair: Integrating Full Volume Experiments and Computational Modeling. *In Emerging Ideas in Mechanical Engineering*, Duke University, Durham, NC, 28–29 May 2026.

IT2. Villacís Núñez, C. N. Towards Rotator Cuff Tendon Digital Twins: A Three-Dimensional Experimental and Computational Approach. Henry Ford Health, Detroit, MI, 10 Feb 2026.

IT3. Villacís Núñez, C. N. Soft tissue mechanics: Full-field strain and displacement maps in an animal model of the rotator cuff. *In II Seminario Internacional de Investigación, Innovación y Transferencia de Tecnología en Ciencias de la Energía y Mecánica at Universidad de las Fuerzas Armadas - ESPE*. Sangolquí, Ecuador, 23 Nov 2023.

IT4. Villacís Núñez, C. N. 3D strain analysis of an animal model of the rotator cuff. *In Orthopaedic Research Laboratories (ORL) Fall 2022 Seminar Series at the University of Michigan*. Ann Arbor, MI, 15 Nov 2022.

IT5. Villacís Núñez, C. N. Tips to apply for scholarships abroad & experIEEEnce of being a woman in engineering. *In Ciclo de Conferencias IEEE WEEK 2021, Rama Estudiantil IEEE – ESPE*. Remote, 20 Jan 2021.

Conference Talks

CT1. Villacís Núñez, C. N., Scheven, U., Bedi, A., Arruda, E. M., Gavini, V., & Huan, X. Toward tear repair optimization using a digital twin of rotator cuff tendon surrogate. *In ASME 2026 International Mechanical Engineering Congress and Exposition (IMECE 2026)*. Vancouver, Canada, 08–12 Nov 2026. [Accepted]

CT2. Villacís Núñez, C. N., Scheven, U., Bedi, A., Arruda, E. M., Gavini, V., & Huan, X. Toward tear repair optimization using a digital twin of rotator cuff tendon surrogate. *In Society of Engineering Science Technical Meeting 2026*. West Lafayette, IN, 11–14 Oct 2026. [Accepted]

CT3. Villacís Núñez, C. N. & Andrade Damasceno, D. Learn science with Newton: Your biomechanics expert. *In Commun-AI-ty: Building Science Communication Skills for Explaining AI in Science*. Ann Arbor, MI, 23–26 Jun 2026. (**Runner up of web design competition**)

CT4. Villacís Núñez, C. N., Wang, Z., Srivastava, S., Scheven, U., Bedi, A., Garikipati, K., & Arruda, E. M. Unraveling rotator cuff tendon tear growth mechanisms with full volume strains and data-driven modeling. *In Michigan Institute for Computational Discovery and Engineering, Ph.D. in Scientific Computing Seminar Series*. Ann Arbor, MI, 11 Mar 2025.

CT5. Villacís Núñez, C. N., Ha, H. J., & Arruda, E. M. Experimental characterization and shear strain maps of mode II failure in an animal model of the rotator cuff. *In ASME 2024 International Mechanical Engineering Congress and Exposition (IMECE 2024)*. Portland, OR, 17–21 Nov 2024.

CT6. Villacís Núñez, C. N., Scheven, U., Bedi, A., & Arruda, E. M. Complex mechanical behavior in intact and torn rotator cuff tendons using full-field magnetic resonance imaging-based strains and data-driven variational system identification. *In ASME 2024 International Mechanical Engineering Congress and Exposition (IMECE 2024)*. Portland, OR, 17–21 Nov 2024.

CT7. Villacís Núñez, C. N., Wang, Z., Srivastava, S., Scheven, U., Bedi, A., Garikipati, K., & Arruda, E. M. Intact and torn rotator cuff behavior using magnetic resonance imaging and variational system identification. *In British Society for Strain Measurement (BSSM) 18th International Conference on Advances in Experimental Mechanics 2024*. Liverpool, England, 03–05 Sep 2024.

CT8. Villacís Núñez, C. N., Scheven, U., Bedi, A., & Arruda, E. M. Full-field strains in high-grade partial-thickness rotator cuff tendon tears reveal shear-based tear progression mechanism. *In The 26th International Congress of Theoretical and Applied Mechanics (ICTAM) 2024*. Daegu, Korea, 25–30 Aug 2024.

CT9. Villacís Núñez, C. N., Scheven, U., Bedi, A., & Arruda, E. M. Full-field strains of high-grade bursal-sided rotator cuff tendon tears. *In 2024 Society for Experimental Mechanics (SEM) Annual Conference*. Vancouver, WA, 03–06 Jun 2024.

CT10. Villacís Núñez, C. N., Scheven, U., Bedi, A., & Arruda, E. M. Full-field strains in an intact and torn rotator cuff animal model using magnetic resonance imaging. *In British Society for Strain Measurement (BSSM) 17th International Conference on Advances in Experimental Mechanics 2023*. Glasgow, Scotland, 30 Aug–01 Sep 2023.

CT11. Villacís Núñez, C. N., Scheven, U., Bedi, A., & Arruda, E. M. Three dimensional strain analysis of intact and artificially torn ovine infraspinatus tendons. *In 2023 Society for Experimental Mechanics (SEM) Annual Conference*. Orlando, FL, 05–08 Jun 2023.

CT12. Villacís Núñez, C. N., Cooper, K. L., & Moore, T. Y. Fusion reinforces metatarsals and facilitates larger body sizes in jerboas (Dipodidae). *In Society for Integrative and Comparative Biology (SICB) 2021 Meeting*. Remote, 03 Jan–28 Feb 2021.

CT13. Narváez, D., **Villacís Núñez, C. N.**, Narváez, C., Vizuete, K., & Arroyo, C. Effect of organic solvents in morphology and mechanical properties of electrospun polyvinylpyrrolidone fibers. *In XIII Congreso de Ciencia & Tecnología*. Jun 2018.

Conference Posters

CP1. Villacís Núñez, C. N., Arruda, E. M., Gavini, V., Huan, X. Toward repair optimization using digital twins of soft tissue. *In Engineering in Cardiovascular Medicine*. Ann Arbor, MI, 06–07 Aug 2026.

CP2. Villacís Núñez, C. N., Srivastava, S., Scheven, U., Bedi, A., Garikipati, K., & Arruda, E. M. Constitutive parameter inference in full volume datasets of intact and torn rotator cuff tendons. *In 10th World Congress of Biomechanics*. Vancouver, Canada, 11–15 Jul 2026.

CP3. Zhang, M., **Villacís Núñez, C. N.**, Yu, R., & Moore, T. Y. Geometrically tuning soft joints to mimic intervertebral joint mechanics. *In Midwest Robotics Workshop*. Chicago, IL, 11–12 Jun 2026.

CP4. Villacís Núñez, C. N., Arruda, E. M., Gavini, V., Huan, X. Digital twin of a rotator cuff tendon surrogate for suture placement optimization. *In U-M Uncertainty Quantification (UQ) Incubator: Building Trustworthy AI for Science and Engineering*. Ann Arbor, MI, 31 May–03 Jun 2026.

CP5. Zhang, M., **Villacís Núñez, C. N.**, Yu, R., & Moore, T. Y. Geometrically tuning soft joints to mimic intervertebral joint mechanics. *In 9th IEEE International Conference on Soft Robotics (RoboSoft 2026)*. Kanazawa, Japan, 07–11 Apr 2026.

CP6. Agilaw, T., Song, Y., Temple, S., Taaki, J., Chen, H., **Villacís Núñez, C. N.**, Liu, X. Evaluating hybrid models for sediment yield prediction and watershed process understanding. *In American Geophysical Union 2025 (AGU25) Annual Meeting*. New Orleans, LA, 15–19 Dec 2025.

CP7. Villacís Núñez, C. N., Srivastava, S., Scheven, U., Bedi, A., Garikipati, K., & Arruda, E. M. Data-driven constitutive parameter inference in full volume datasets of rotator cuff tendons. *In 20th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering*. Barcelona, Spain, 03–05 Sep 2025.

CP8. Villacís Núñez, C. N., Scheven, U., Bedi, A., & Arruda, E. M. Shear strains as a tear growth mechanism for high-grade partial-thickness rotator cuff tendon tears. *In Summer Biomechanics, Bioengineering, and Biotransport (SB3C) 2024 Conference*. Lake Geneva, WI, 11–14 Jun 2024.

CP9. Villacís Núñez, C. N., Wang, Z., Srivastava, S., Scheven, U., Bedi, A., Garikipati, K., & Arruda, E. M. Rotator cuff constitutive response with full-field displacements and data-driven variational system identification. *In Scientific Foundation Models MICDE Conference 2024.* Ann Arbor, MI, 02–03 Apr 2024.

CP10. Villacís Núñez, C. N., Scheven, U., Bedi, A., & Arruda, E. M. Towards understanding tear progression of high-grade bursal-sided rotator cuff tendon tears with full-field strains. *In Orthopaedic Research Society (ORS) 2024 Annual Meeting*. Long Beach, CA, 02–06 Feb 2024.

CP11. Villacís Núñez, C. N., Scheven, U., Bedi, A., Arruda, E. M. Full-field strain analysis in an animal model of the rotator cuff with partial-thickness tears. *In Seventh Annual Musculoskeletal Health Symposium from the Michigan Integrative Musculoskeletal Health Core Center*. Ann Arbor, MI, May 11 2023.

CP12. Villacís Núñez, C. N., Scheven, U., Bedi, A., & Arruda, E. M. Three-dimensional strains in an animal model of the rotator cuff with artificially created partial-thickness tears. *In Orthopaedic Research Society (ORS) 2023 Annual Meeting*. Dallas, Texas, 10–14 Feb 2023.

CP13. Villacis, N., Li, D-R., Shih, A. Needle computer numerical control microgrinding with vision-guided method. *In Manufacturing Science and Engineering Conference (MSEC) 2019*. Erie, PA, 10–14 Jun 2019.

Fellowships, Honors, and Awards

BME UNITE Future Faculty Fellow and Travel Award, BME UNITE, $1,350 2026
Designed to showcase faculty candidates from diverse and underrepresented backgrounds in BME

ASB NextGen Biomechanics Travel Award, De Luca Foundation, $2,000 2026
Intended to support individuals from different backgrounds who are committed to advancing biomechanics

Travel Scholarship for Data and Intensive Research with Rigor and Reproducibility Program 2026
Awarded to researchers dedicated to advancing rigorous scientific practices at their home institution

Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship, Schmidt Sciences, $164,500 2025
Granted to an outstanding researcher to pivot their direction into AI methods

ProQuest Distinguished Dissertation Award Nominee, University of Michigan 2025
Nominated for representing the best scholarly work in Mechanical Engineering dissertations

Robert M. Caddell Memorial Award for Research, University of Michigan**,** $2,500 2025
Granted to a Faculty/Grad Student team that have made significant contributions to research in materials

Rackham Professional Development Grant, University of Michigan, $267 2024
Intended to support short-term experiences for the development of professional skills

MICDE Fellowship, University of Michigan, $5,000 2024
Presented to graduate students whose projects advance scientific computing techniques and practices

2024 ASME Applied Mechanics Division Travel Award, IMECE 2024, $1,000 2024
Intended to defray conference costs for speakers with innovative thesis accepted at IMECE

Rising Stars in Mechanical Engineering, Carnegie Mellon University 2024
Awarded to exceptional women graduate students and postdocs who are considering careers in academia

Runner-up at Young Stress Analyst Competition, BSSM 2024 2024
Presented at the BSSM 2024 Conference for excellence in early career research

Thomas J. R. Hughes Fellowship, ICTAM 2024, $2,000 2024
Awarded at the ICTAM 2024 to early career researchers with potential for future excellence in mechanics

Rackham Predoctoral Fellowship, University of Michigan, $38,970 2024
Intended to support students working on dissertations that are unusually creative, ambitious, and impactful

Ivor K. McIvor Award, University of Michigan, $3,000 2024
Awarded to a senior ME graduate student who has demonstrated excellence in Applied Mechanics

Alexander Azarkhin Fellowship, University of Michigan, $2,500 2023
Presented annually to an ME Ph.D. student who has an interest in Applied Mechanics

Rackham Conference Travel Grant, University of Michigan, $3,500 in total 2022, 2023, 2024
Intended to provide academic opportunities for Rackham graduate students

NSF Fellowship for Mechanistic Machine Learning and Digital Twins for Computational Science, Engineering & Technology 2021 Conference, NSF 2021
Covered the remote registration fee for the 2021 MMLDT-CSET Conference and a short course

Rackham International Student Fellowship, University of Michigan, $10,000 2019
Presented to outstanding international students with a strong academic record and professional promise

Graduate Student Co-Curricular Activities Funding, University of Michigan, $815 2019
Intended to provide financial support for professional development activities

Charles Vest Award, University of Michigan, $10,000 2018
Awarded by UofM alumnus and Professor Charles Vest

Rice-Cullimore Scholarship, ASME, $3,000 2018
Presented by ASME to a foreign student intending to do graduate work in ME in the United States

Fulbright Foreign Student Program, Fulbright Commission Ecuador, $30,000 2018
Awarded to outstanding Ecuadorian citizens for their graduate studies in the United States

Programa de Becas Internacionales de Posgrado, Senescyt Ecuador, $82,000 2018
Intended to encourage outstanding Ecuadorian citizens to study abroad

Grant Writing Experience

Community Initiatives Funds, Schmidt Sciences 2026
- Role: Co-lead and writer
- Proposal title: “The Synergy Hub - Healthcare Edition: Piloting a Co-Design Roadmap for Translational AI in Science”

Science Communication Micro-Awards, University of Michigan 2022, 2023
- Role: Lead and writer
- Proposal title: “Beyond Machines: How Engineers Use Digital Twins to Improve Healthcare”

MiMHC Pilot Grant Application, University of Michigan 2022, 2023
- Role: Writer and co-investigator
- 2023 Proposal: “Full-Field Strain Distributions and Concentrations in a Torn and Repaired Animal Model of the Rotator Cuff”
- 2022 Proposal: “Investigation of tendon strains and tissue damage in a model for human supraspinatus tears, using the displacement encoding MRI method *MR-u*.”

Mentoring Experience

Ryan Yu, B.S. in Data Science, anticipated graduation in 2029 2025–Present
- Project: Shape identification through diffusion models
- Achievements: Poster presentation at Midwest Robotics Workshop 2026

Hyun Jun Ha, B.S. in Mechanical Engineering 2024–Present
- Project 1: Suture placement optimization in simplified models of stiff fascicles and compliant matrix
- Project 2: Mode II tear propagation in ovine rotator cuff tendons
- Achievements: Oral presentation at IMECE 2024, continued to a Ph.D. program in Mechanical Engineering

Dylan Andresino, B.S. in Mechanical Engineering 2023
- Project: Exploratory fixture design for ovine rotator cuff tendons

Andrew P. Ray, B.S. in Materials Science 2020–2021
- Project: Dynamical analysis of jerboa bones subjected to impact to understand evolutionary trait effects
- Achievements: Co-author in publication at Proceedings of the Royal Society B

Gabriel Melo, B.S. in Mechanical Engineering 2017–2018
- Project 1: Electrospinning-based wound dressings made of dragon's blood to treat chronic injuries
- Project 2: Fabrication of 3D printed hand prosthesis for partial hand amputation
- Achievements: Two oral presentations and conference proceedings at "XIII Congreso de Ciencia y Tecnología - ESPE", and one conference proceeding at "Revista Ciencia"

Diego Narváez, B.S. in Mechanical Engineering 2017–2018
- Project: Effect of organic solvents on morphology and mechanical properties of electrospun fibers
- Achievements: Oral presentation at "XIII Congreso de Ciencia y Tecnología - ESPE"

Diego Herrera and **Gabriela Granja**, B.S. in Mechanical Engineering 2017–2018
- Project: Fabrication of wound dressings made of chitosan and dragon's blood
- Achievements: Both completed their thesis with me providing technical support

Germán Moncayo, B.S. in Mechanical Engineering 2017–2018
- Project: Mechanical properties of Ecuadorian balsa wood (Ochroma Pyramidale)
- Achievements: Oral presentation at "XIII Congreso de Ciencia y Tecnología - ESPE"

Teaching Experience

Guest Lecturer – CSE 598-002: AI for Science 2025
Computer Science and Engineering, University of Michigan, Ann Arbor, MI
- Led two interactive sessions to guide students on AI/ML approach selection for constitutive model inference with full volume datasets

Graduate Student Instructor – ME 201: Introduction to Solid Mechanics 2021
Mechanical Engineering, University of Michigan, Ann Arbor, MI
- Led two discussion sections, helping students develop problem-solving skills through detailed homework exercises
- Taught students to construct free body diagrams, applying and reinforcing concepts learned in class, such as force and moment equilibrium
- Graded exams for more than 80 students and provided feedback addressing student concerns

Hands-on Instructor – "Reportes de Ingeniería con LaTeX" 2021
IEEE ESPE, Remote webinar
- Taught high school and college students to use Overleaf, including figure and table creation, the use of templates, and reference management

Hands-on Instructor – MATLAB and LaTeX 2021
Mechanical Engineering Graduate Council, University of Michigan, Ann Arbor, MI
- Guided graduate students through interactive activities and coding problems to learn MATLAB and LaTeX, providing them with ready-to-use templates

Leadership and Service

Data Science & AI Summit 2025 Poster Judge, University of Michigan 2025

U-M Leadership AI Summit 2025 Volunteer, University of Michigan 2025

MIDAS Carpentries, University of Michigan

Neuroscience Exploration with AI Co-Chair	2025–Present
Physics-Informed Neural Networks Co-Chair	2025–2026
UQ Incubator Volunteer	2026

Mechanical Engineering Graduate Council (MEGC), University of Michigan

Professional Development Co-Chair	2023–2024
Seminars Chair	2022–2023
President	2021–2022
Workshops Chair	2020–2021
DEI Co-Chair	2019–2020

Research Fundamentals Exam Practice Reviewer, University of Michigan 2022, 2023

B-SURE Program Panelist, American Society of Biomechanics 2023

ME Undergraduate Symposium Poster Judge, University of Michigan 2020, 2022, 2023

Grad School 101 Panelist, University of Michigan 2022

Graduate Student Advisory Council (GSAC) Representative, University of Michigan 2021–2022

Tau Beta Pi DEI Committee Co-Chair, University of Michigan 2021

Peer Review

Ad hoc reviewer for:

- *Acta Biomaterialia*, 1 manuscript
- *International Journal of Solids and Structures*, 1 manuscript
- *Computer Methods in Biomechanics and Biomedical Engineering*, 1 manuscript

Professional Affiliations

Biomedical Engineering Society	2026–Present
Sigma Xi Honor Society	2025–Present
American Society of Mechanical Engineers	2024–Present
American Society of Biomechanics	2024–Present
National Postdoctoral Association	2025–Present

Relevant Skills

Experimental Mechanics: Full volume analysis, material characterization, soft tissue mechanical testing
CAD and Design: Autodesk Inventor, Solidworks, GrabCAD, Meshmixer, custom fixture design, 3D printing
Finite Element Analysis and Modeling: Python, FEniCS, Fortran, Abaqus, COMSOL, Hypermesh
Data Analysis and Scientific Computing: MATLAB, Paraview
Machine Learning Tools: PyTorch, scikit-learn
Machine Learning Methods: neural ODEs, diffusion models